Reinforcement learning from human feedback

Reward models, the KL penalty and PPO

Carlos Cotrini

ETH Zurich

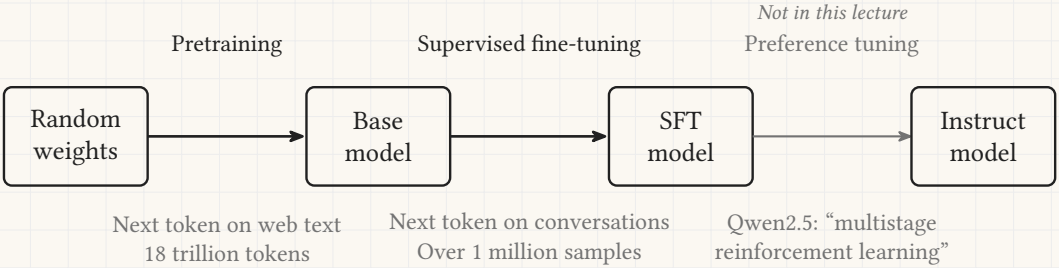
20 October 2026

1

Section 1

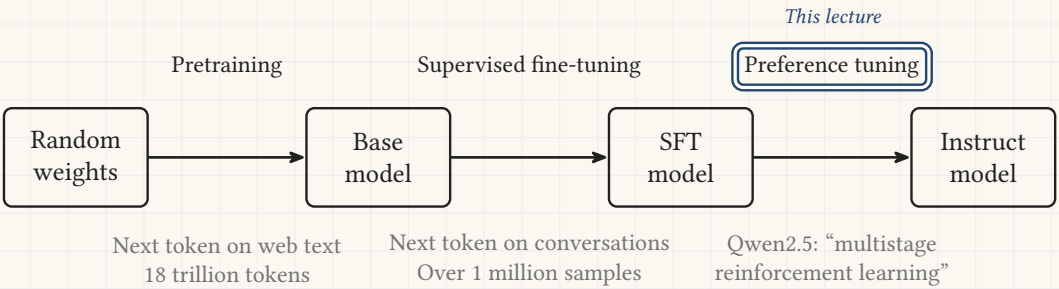
Learning from comparisons

One more training stage



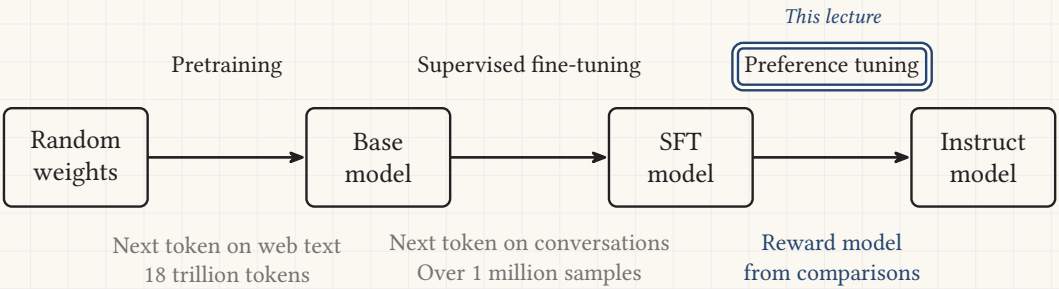
Qwen2.5 Technical Report, arXiv:2412.15115

One more training stage



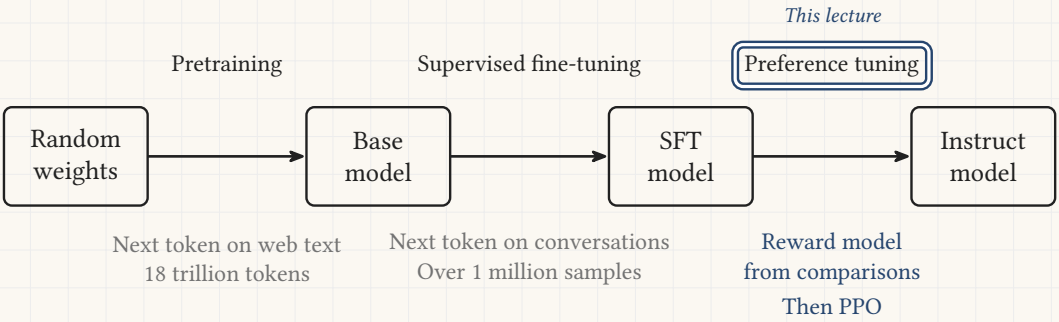
Qwen2.5 Technical Report, arXiv:2412.15115

One more training stage



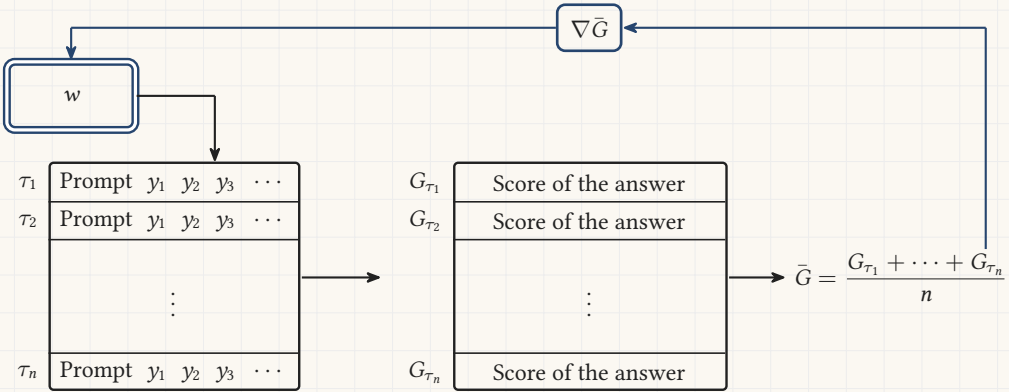
Qwen2.5 Technical Report, arXiv:2412.15115

One more training stage

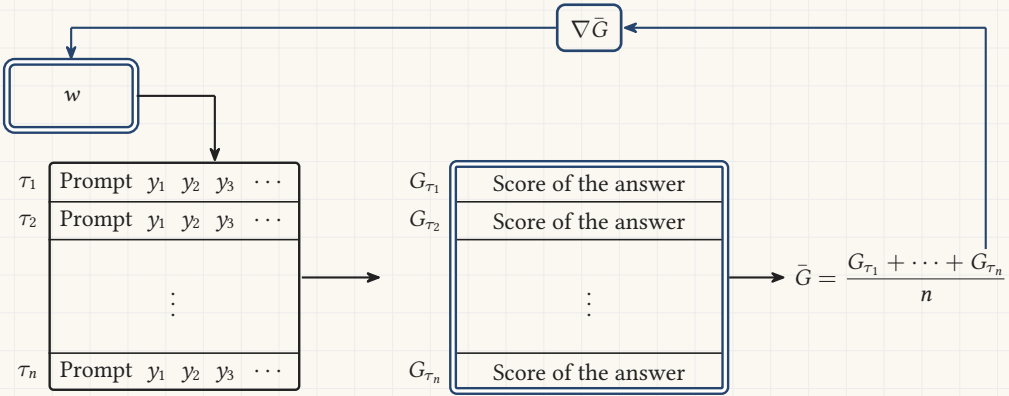


Qwen2.5 Technical Report, arXiv:2412.15115

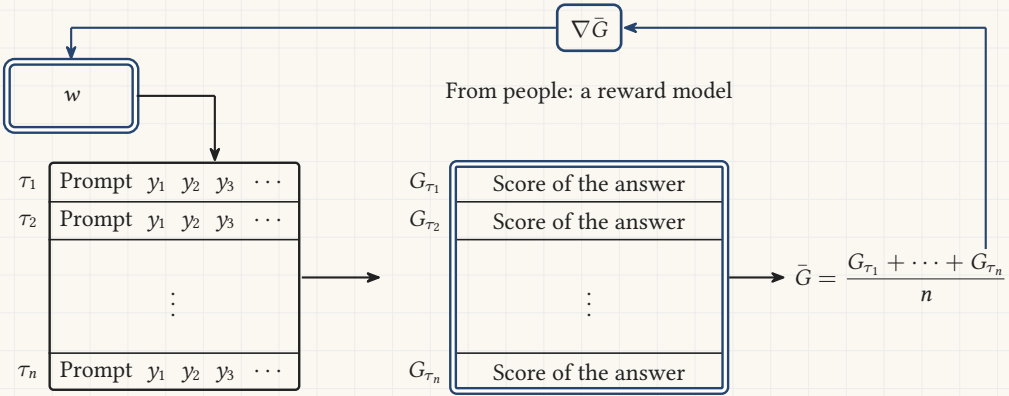
The same loop for a language model



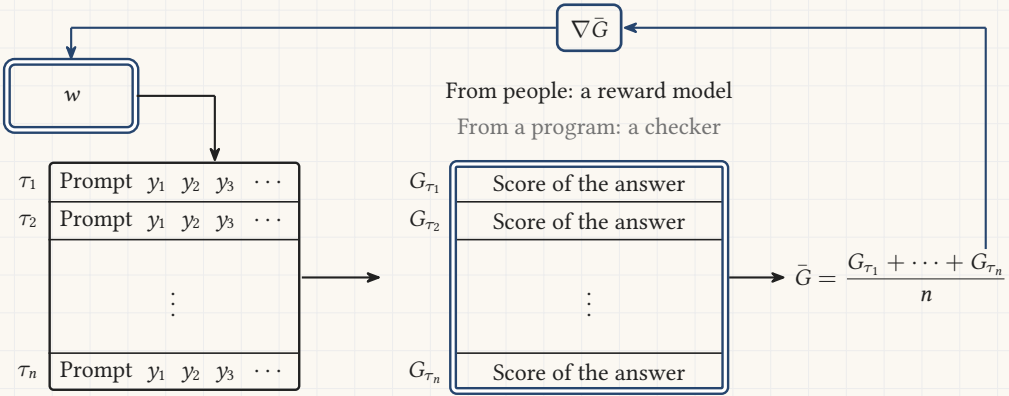
The same loop for a language model



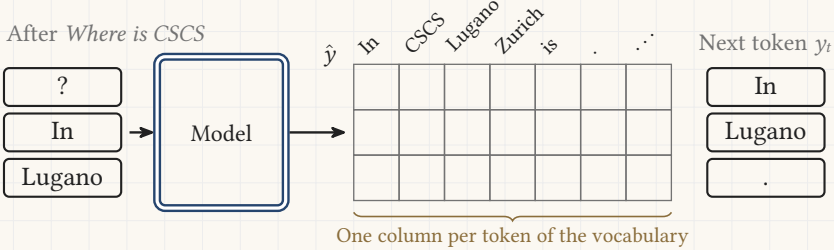
The same loop for a language model



The same loop for a language model

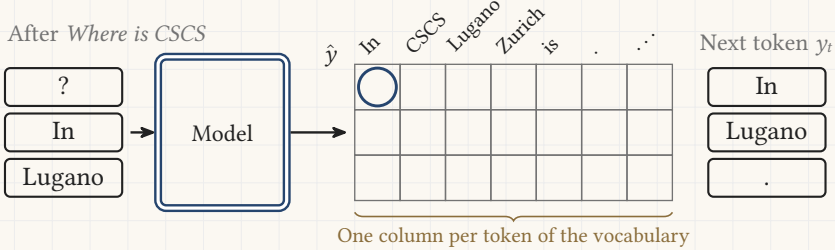


The SFT loss reads one answer



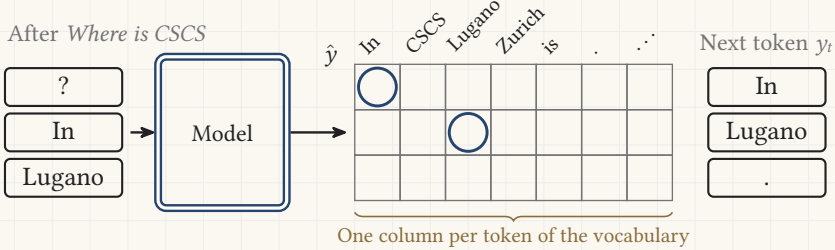
The SFT loss reads one answer

$$\log \hat{y}_{1,\text{In}}$$



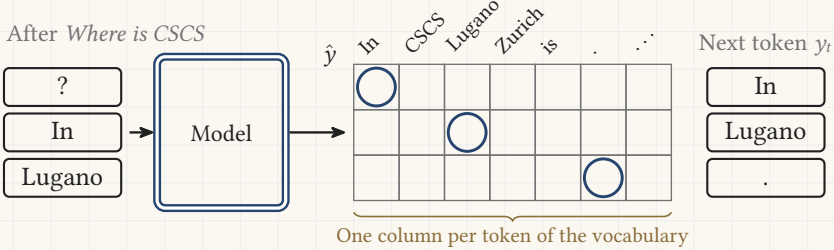
The SFT loss reads one answer

$$\log \hat{y}_{1,\text{In}} + \log \hat{y}_{2,\text{Lugano}}$$



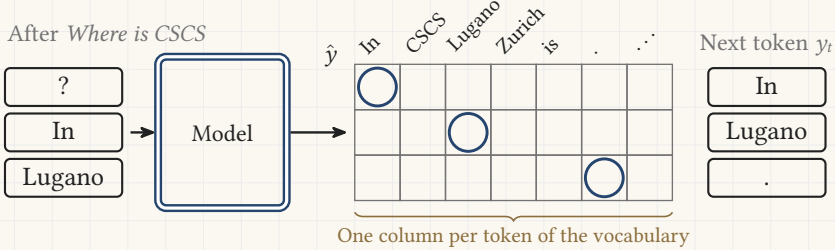
The SFT loss reads one answer

$$\log \hat{y}_{1,\text{In}} + \log \hat{y}_{2,\text{Lugano}} + \log \hat{y}_{3,.}$$



The SFT loss reads one answer

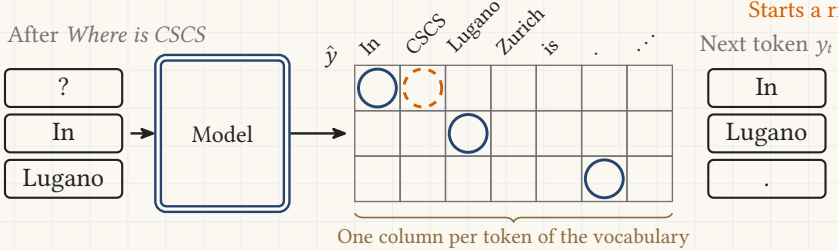
$$\text{SFT loss} \longrightarrow \bar{l} = -\frac{1}{3}\underbrace{\left(\log \hat{y}_{1,\text{In}} + \log \hat{y}_{2,\text{Lugano}} + \log \hat{y}_{3,.}\right)}_{\text{Circled probabilities}}$$



The SFT loss reads one answer

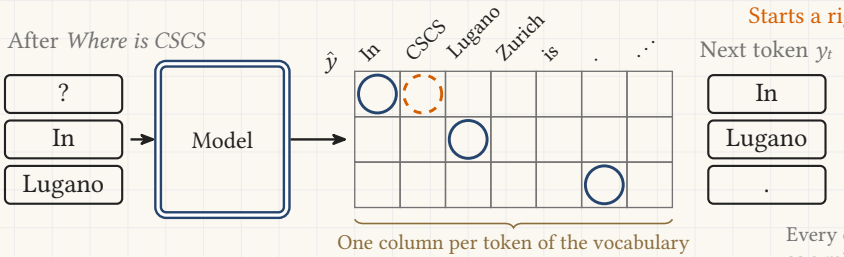
SFT loss $\longrightarrow \bar{l} = -\frac{1}{3}\underbrace{\left(\log \hat{y}_{1,\text{In}} + \log \hat{y}_{2,\text{Lugano}} + \log \hat{y}_{3,.}\right)}_{\text{Circled probabilities}}$

Pushed down: $\hat{y}_{1,\text{CSCS}}$
Starts a right answer



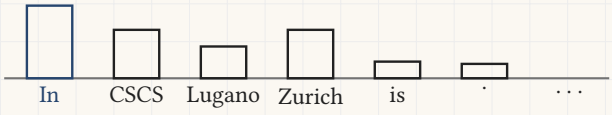
The SFT loss reads one answer

SFT loss $\longrightarrow \bar{l} = -\frac{1}{3}\underbrace{\left(\log \hat{y}_{1,\text{In}} + \log \hat{y}_{2,\text{Lugano}} + \log \hat{y}_{3,.}\right)}_{\text{Circled probabilities}}$



A gradient step may lower other correct tokens

After *Where is CSCS ?* ($\hat{y}_{1,k}$)

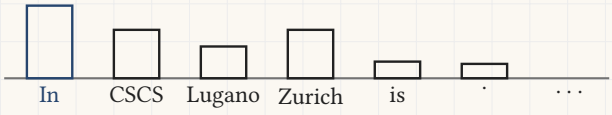


A gradient step may lower other correct tokens

Row 1's cross-entropy → $\partial\,(-\log \hat{y}_{1,\text{In}})$

Logit of token k → $\partial z_{1,k}$

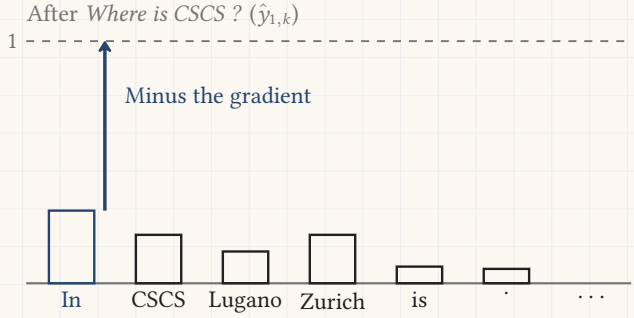
After *Where is CSCS ?* ($\hat{y}_{1,k}$)



A gradient step may lower other correct tokens

Row 1's cross-entropy → $\frac{\partial\,(-\log \hat{y}_{1,\mathrm{In}})}{\partial z_{1,k}} = \begin{cases} \hat{y}_{1,k} - 1 & k = \mathrm{In} \end{cases}$

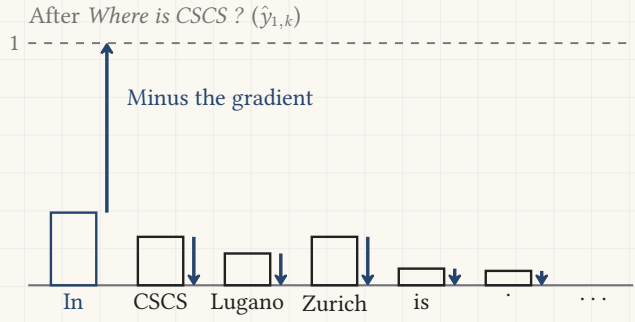
Logit of token k →



A gradient step may lower other correct tokens

Row 1's cross-entropy → $\frac{\partial\,(-\log \hat{y}_{1,\text{In}})}{\partial z_{1,k}} = \begin{cases} \hat{y}_{1,k} - 1 & k = \text{In} \\ \hat{y}_{1,k} & \text{every other } k \end{cases}$

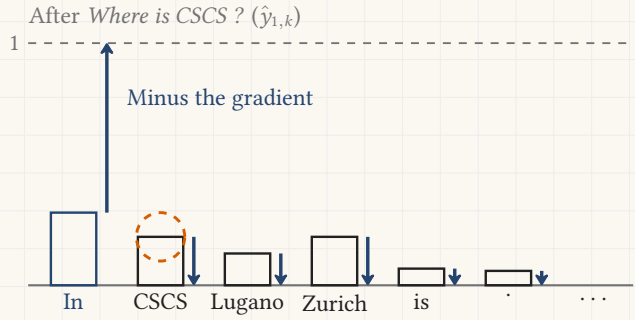
Logit of token k →



A gradient step may lower other correct tokens

Row 1's cross-entropy → $\dfrac{\partial\,(-\log \hat{y}_{1,\text{In}})}{\partial z_{1,k}} = \begin{cases} \hat{y}_{1,k} - 1 & k = \text{In} \\ \hat{y}_{1,k} & \text{every other } k \end{cases}$

Logit of token k →



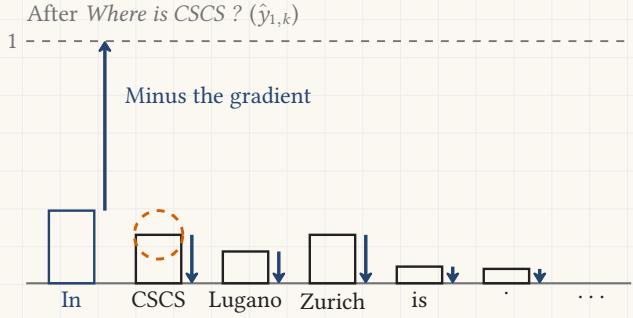
A right answer

CSCS | is | in | Lugano | .

A gradient step may lower other correct tokens

Row 1's cross-entropy → $\dfrac{\partial\,(-\log \hat{y}_{1,\text{In}})}{\partial z_{1,k}} = \begin{cases} \hat{y}_{1,k} - 1 & k = \text{In} \\ \hat{y}_{1,k} & \text{every other } k \end{cases}$

Logit of token k →



A right answer

CSCS | is | in | Lugano | .

A wrong answer

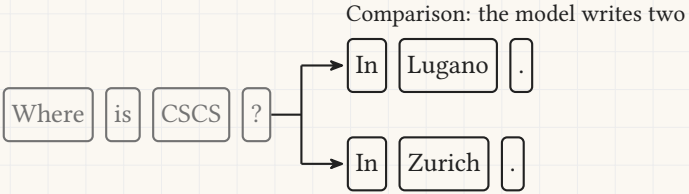
Zurich | .

The step lowers CSCS exactly as it lowers Zurich.

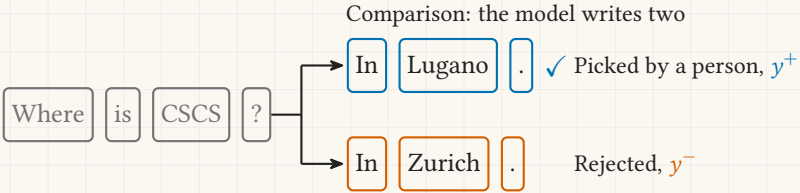
A comparison in place of a demonstration



A comparison in place of a demonstration

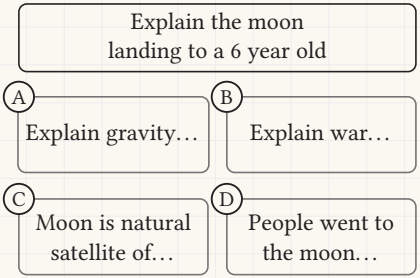


A comparison in place of a demonstration



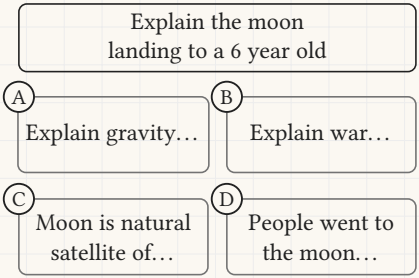
Rankings become pairs

InstructGPT: Ouyang et al. 2022, Fig. 2



Rankings become pairs

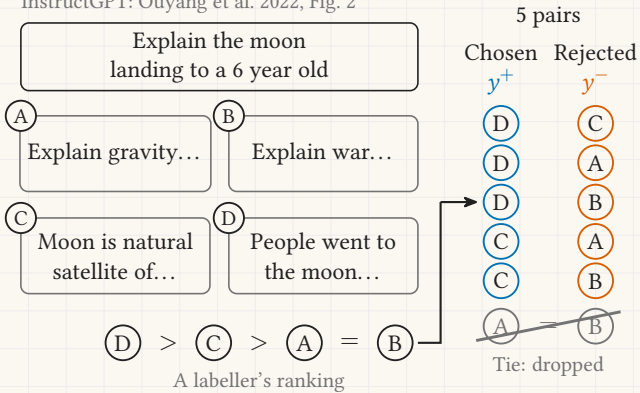
InstructGPT: Ouyang et al. 2022, Fig. 2



A labeller's ranking

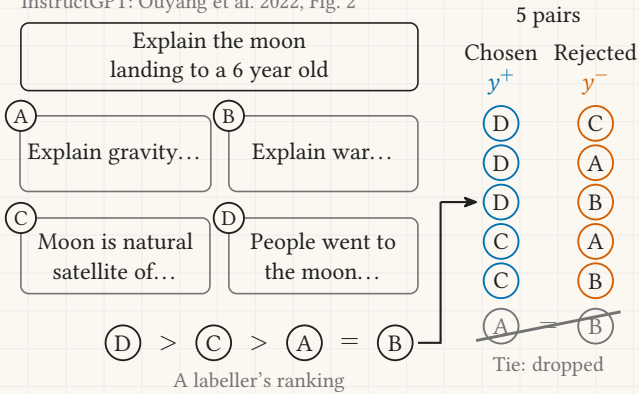
Rankings become pairs

InstructGPT: Ouyang et al. 2022, Fig. 2



Rankings become pairs

InstructGPT: Ouyang et al. 2022, Fig. 2

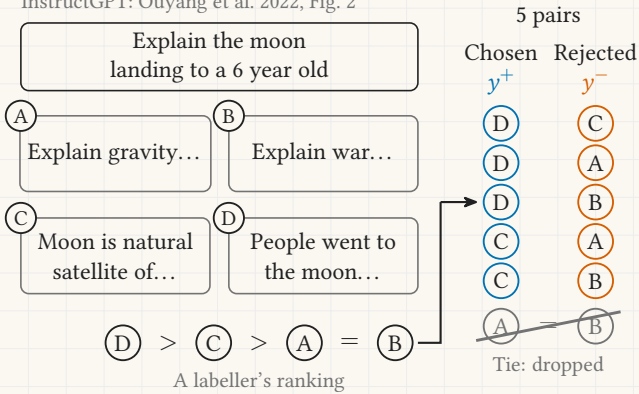


4 to 9 answers per prompt Sec. 3.5

$\binom{4}{2} = 6$ to $\binom{9}{2} = 36$ pairs

Rankings become pairs

InstructGPT: Ouyang et al. 2022, Fig. 2



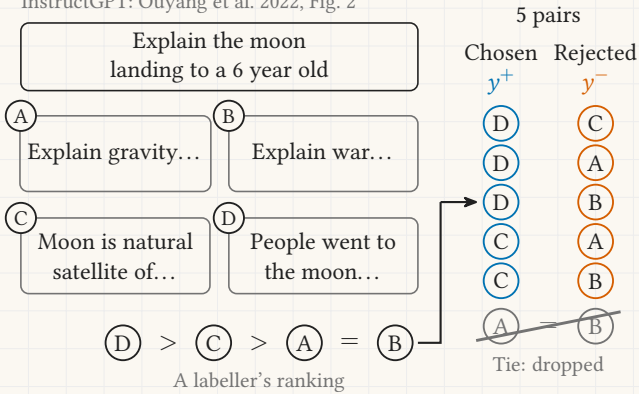
4 to 9 answers per prompt Sec. 3.5

$\binom{4}{2} = 6$ to $\binom{9}{2} = 36$ pairs

About 40 labellers Sec. 3.4

Rankings become pairs

InstructGPT: Ouyang et al. 2022, Fig. 2



4 to 9 answers per prompt Sec. 3.5

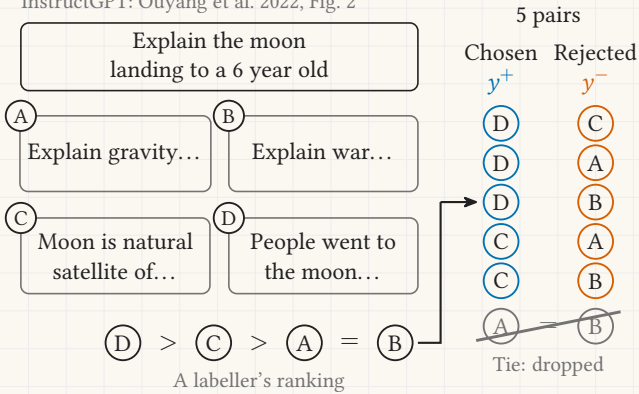
$\binom{4}{2} = 6$ to $\binom{9}{2} = 36$ pairs

About 40 labellers Sec. 3.4

33,207 ranked prompts Table 6

Rankings become pairs

InstructGPT: Ouyang et al. 2022, Fig. 2



4 to 9 answers per prompt — Sec. 3.5

$\binom{4}{2} = 6$ to $\binom{9}{2} = 36$ pairs

About 40 labellers — Sec. 3.4

33,207 ranked prompts — Table 6

Labeller agreement — Sec. 3.4

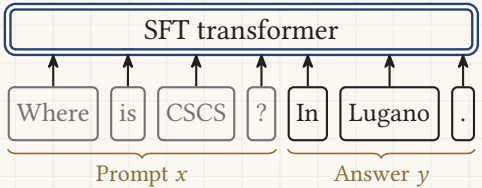
$72.6 \pm 1.5\%$

2

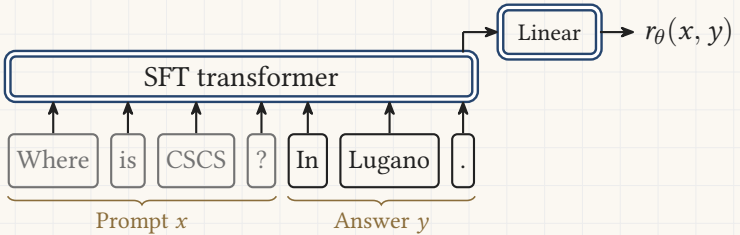
Section 2

The reward model

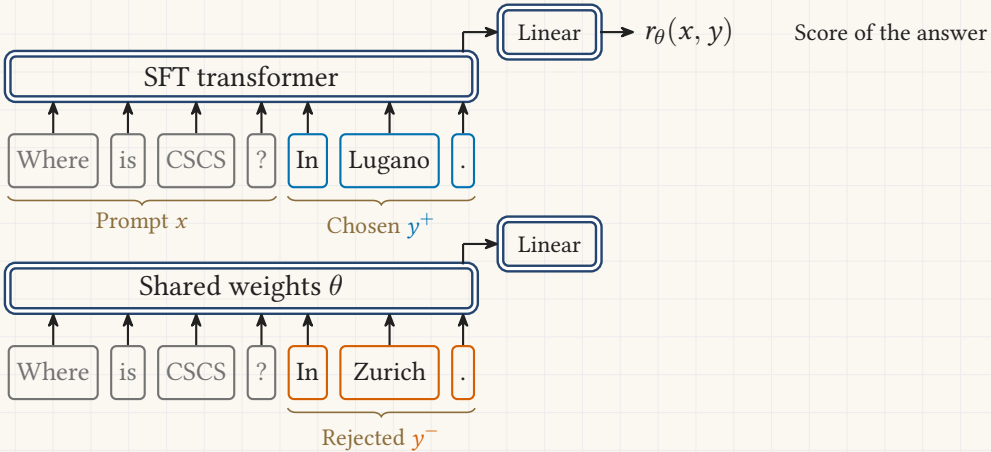
The reward model



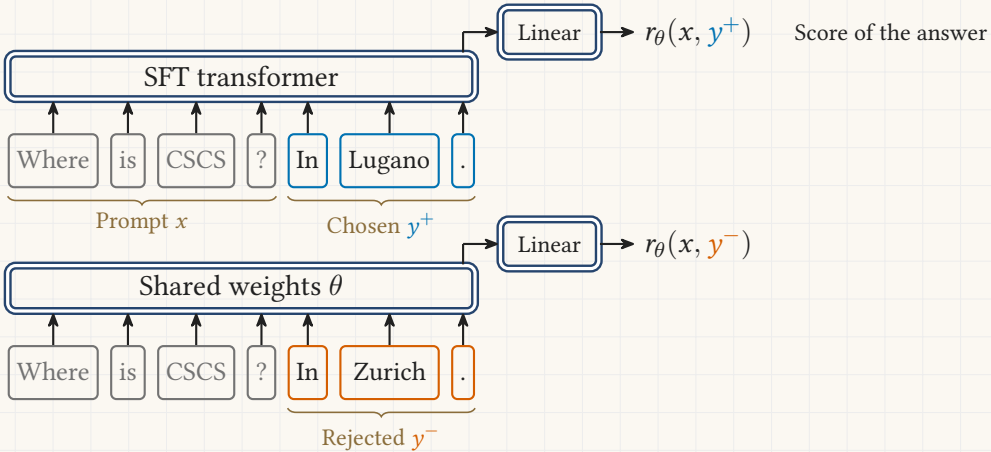
The reward model



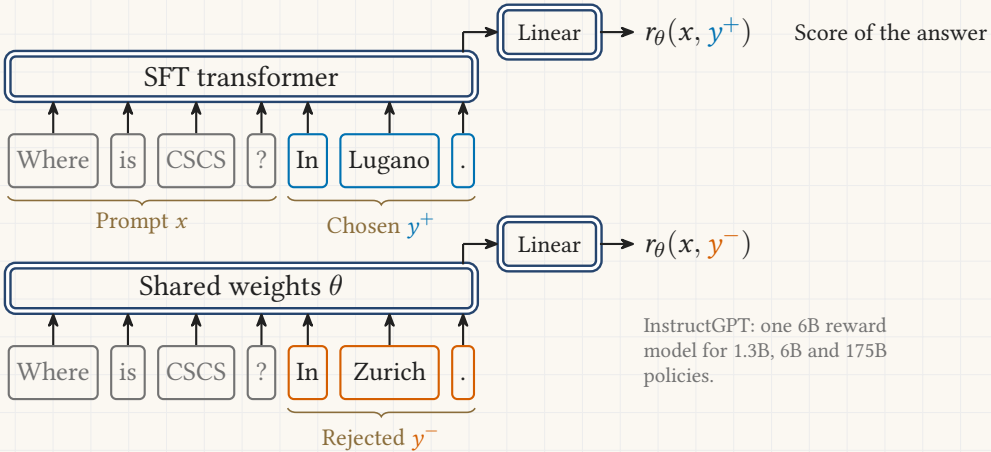
The reward model



The reward model

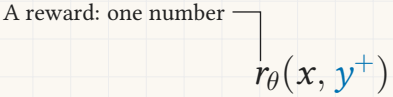


The reward model

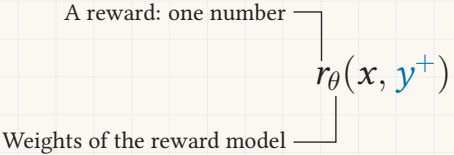


InstructGPT: one 6B reward model for 1.3B, 6B and 175B policies.

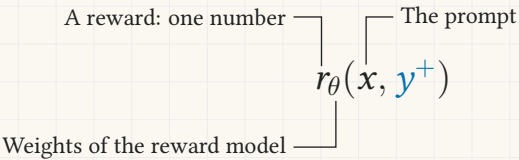
The notation $r_\theta(x, y^+)$



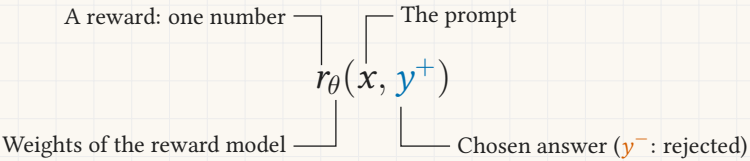
The notation $r_\theta(x, y^+)$



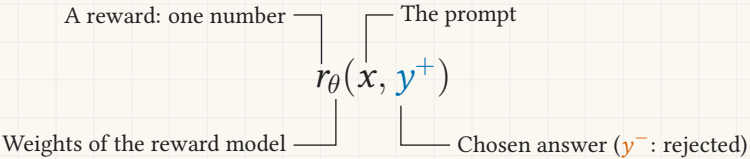
The notation $r_\theta(x, y^+)$



The notation $r_\theta(x, y^+)$

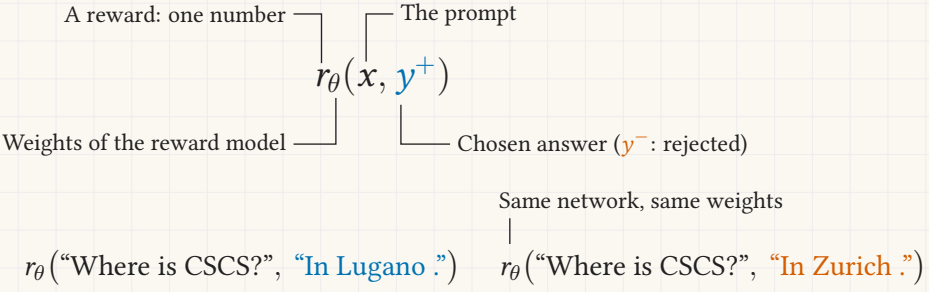


The notation $r_\theta(x, y^+)$

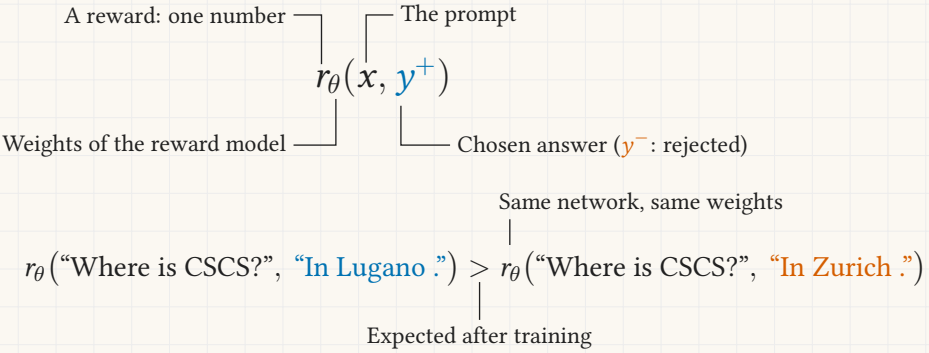


r_θ(“Where is CSCS?”, “In Lugano .”)

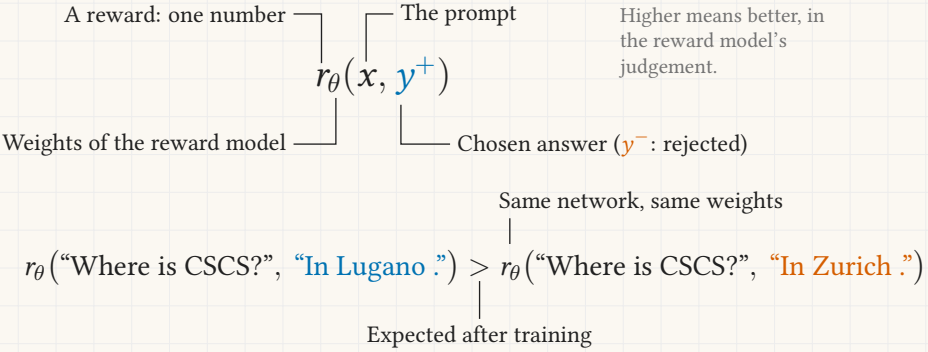
The notation $r_\theta(x, y^+)$



The notation $r_\theta(x, y^+)$



The notation $r_\theta(x, y^+)$



The Bradley-Terry model

$$P\underbrace{\left(y^{+} \succ y^{-}\right)}_{\text{Chosen over rejected}} \mid x\Big)$$

The Bradley-Terry model

$$P\underbrace{\left(y^+ \succ y^-\right)}_{\text{Chosen over rejected}} \mid x\right) = \frac{e^{r_\theta(x,y^+)}}{e^{r_\theta(x,y^+)} + e^{r_\theta(x,y^-)}}$$

Bradley and Terry 1952

The Bradley-Terry model

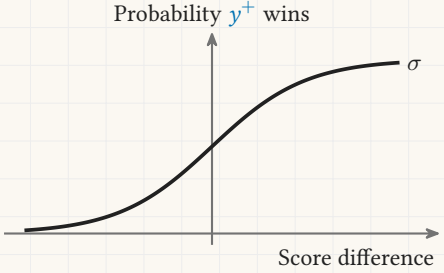
$$P\underbrace{\left(y^+ \succ y^-\right)}_{\text{Chosen over rejected}} \mid x\right) = \frac{e^{r_\theta(x,y^+)}}{e^{r_\theta(x,y^+)} + e^{r_\theta(x,y^-)}} = \sigma\underbrace{\left(r_\theta(x, y^+) - r_\theta(x, y^-)\right)}_{\text{Score difference}}$$

Bradley and Terry 1952

Christiano et al. 2017, eq. (1)

The Bradley-Terry model

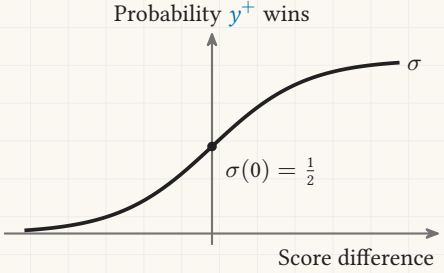
$$P\underbrace{\left(y^+ \succ y^-\right)}_{\text{Chosen over rejected}} \mid x\right) = \frac{e^{r_\theta(x,y^+)}}{e^{r_\theta(x,y^+)} + e^{r_\theta(x,y^-)}} = \sigma\underbrace{\left(r_\theta(x, y^+) - r_\theta(x, y^-)\right)}_{\text{Score difference}}$$



Bradley and Terry 1952
Christiano et al. 2017, eq. (1)

The Bradley-Terry model

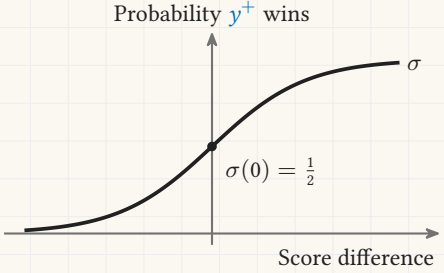
$$P\big(\underbrace{y^+ \succ y^-}_{\text{Chosen over rejected}} \mid x\big) = \frac{e^{r_\theta(x,y^+)}}{e^{r_\theta(x,y^+)} + e^{r_\theta(x,y^-)}} = \sigma\big(\underbrace{r_\theta(x, y^+) - r_\theta(x, y^-)}_{\text{Score difference}}\big)$$



Bradley and Terry 1952
Christiano et al. 2017, eq. (1)

The Bradley-Terry model

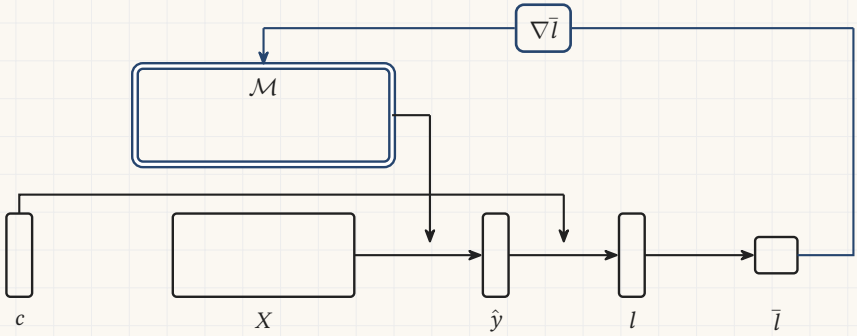
$$P\big(\underbrace{y^+ \succ y^-}_{\text{Chosen over rejected}} \mid x\big) = \frac{e^{r_\theta(x,y^+)}}{e^{r_\theta(x,y^+)} + e^{r_\theta(x,y^-)}} = \sigma\big(\underbrace{r_\theta(x, y^+) - r_\theta(x, y^-)}_{\text{Score difference}}\big)$$



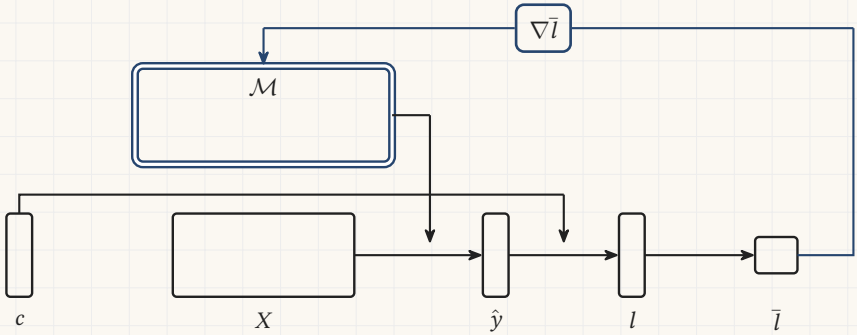
Bradley and Terry 1952
Christiano et al. 2017, eq. (1)

The model behind Elo ratings in chess.

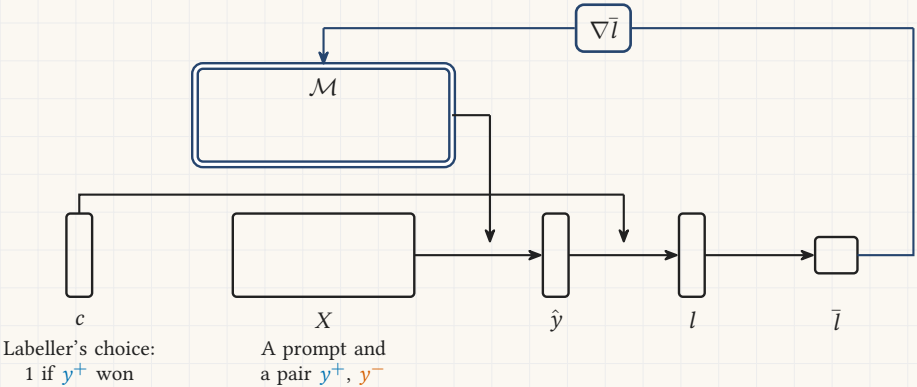
Training the reward model



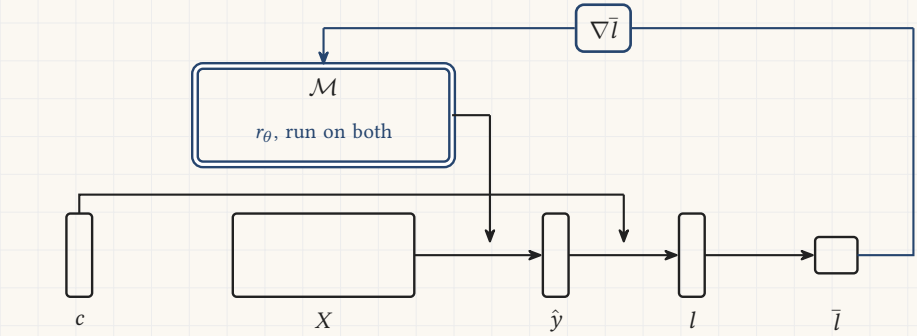
Training the reward model



Training the reward model

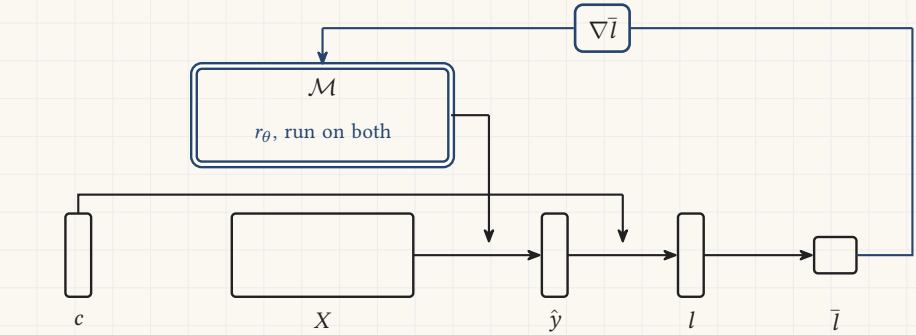


Training the reward model



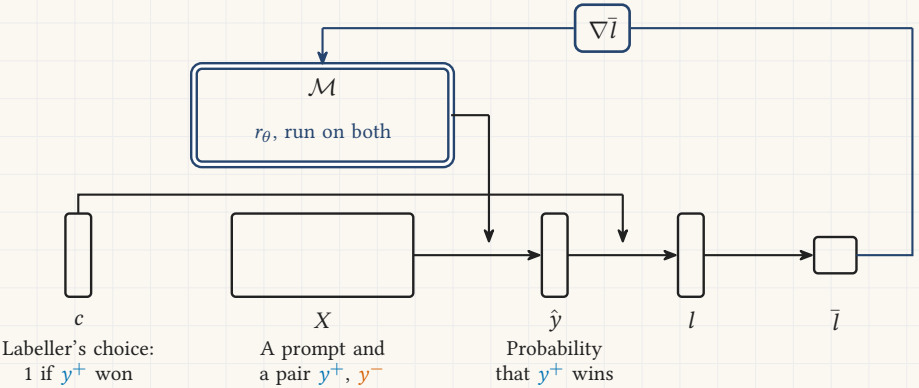
Training the reward model

$$\hat{y} = \sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big)$$



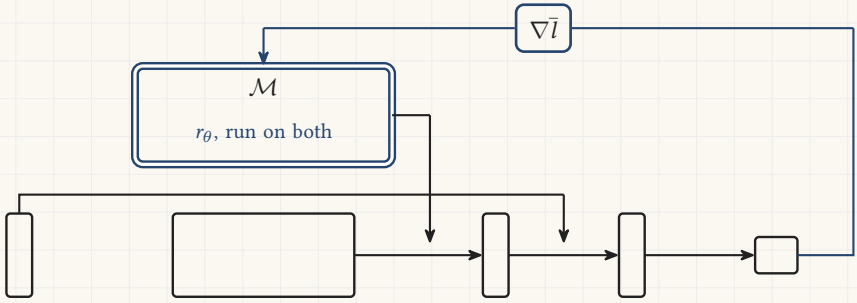
Training the reward model

$$\hat{y} = \sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big) \qquad l = -\log \hat{y}$$



Training the reward model

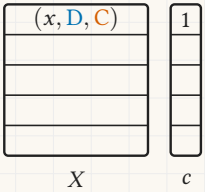
$$\hat{y} = \sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big) \qquad l = -\log \hat{y}$$



The loss over one ranking

$$\underbrace{\log \sigma\left(r_\theta(x, y^+) - r_\theta(x, y^-)\right)}_{\text{One pair}}$$



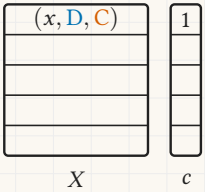


One example per pair

The loss over one ranking

$$\underbrace{\log \sigma\left(r_\theta(x, y^+) - r_\theta(x, y^-)\right)}_{\text{One pair}}$$



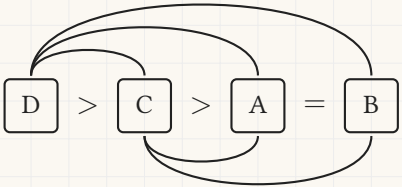


One example per pair

Winner listed first: $c = 1$

The loss over one ranking

$$\underset{\text{pairs of } x}{\text{avg}} \underbrace{\log \sigma\left(r_\theta(x, y^+) - r_\theta(x, y^-)\right)}_{\text{One pair}}$$



5 pairs

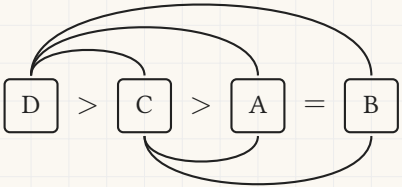
X	c
(x, D, C)	1
(x, D, A)	1
(x, D, B)	1
(x, C, A)	1
(x, C, B)	1

One example per pair

Winner listed first: $c = 1$

The loss over one ranking

$$\text{loss}(\theta) = -\operatorname*{avg}_{\text{prompts } x}\left[\operatorname*{avg}_{\text{pairs of } x}\underbrace{\log\sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big)}_{\text{One pair}}\right]$$



5 pairs

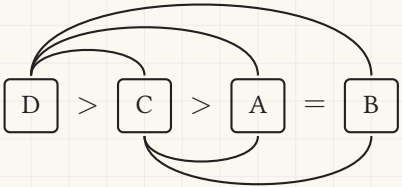
X	c
(x, D, C)	1
(x, D, A)	1
(x, D, B)	1
(x, C, A)	1
(x, C, B)	1

One example per pair

Winner listed first: $c = 1$

The loss over one ranking

$$\text{loss}(\theta) = -\operatorname*{avg}_{\text{prompts } x}\left[\operatorname*{avg}_{\text{pairs of } x}\underbrace{\log\sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big)}_{\text{One pair}}\right]$$



5 pairs

X	c
(x, D, C)	1
(x, D, A)	1
(x, D, B)	1
(x, C, A)	1
(x, C, B)	1

One example per pair

Winner listed first: $c = 1$

All pairs of one prompt in one batch element; one epoch.

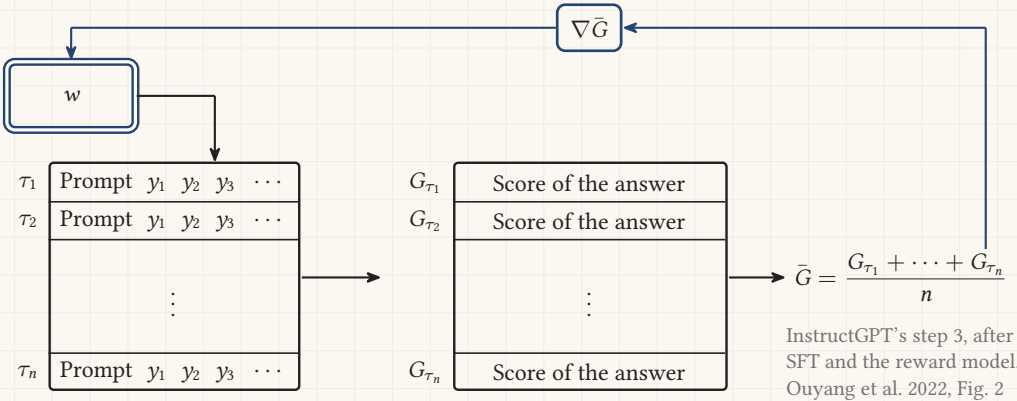
Ouyang et al. 2022, eq. (1), Fig. 2

3

Section 3

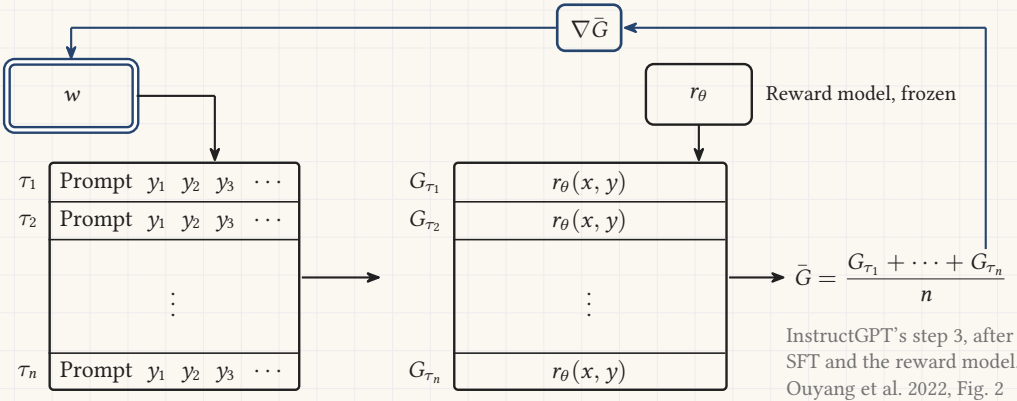
The KL-regularised objective

Reinforcement learning against the reward model



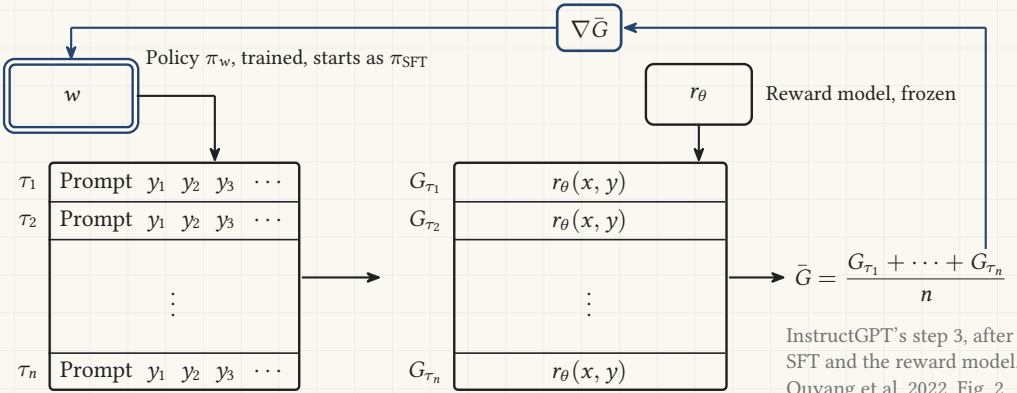
InstructGPT's step 3, after SFT and the reward model. Ouyang et al. 2022, Fig. 2

Reinforcement learning against the reward model



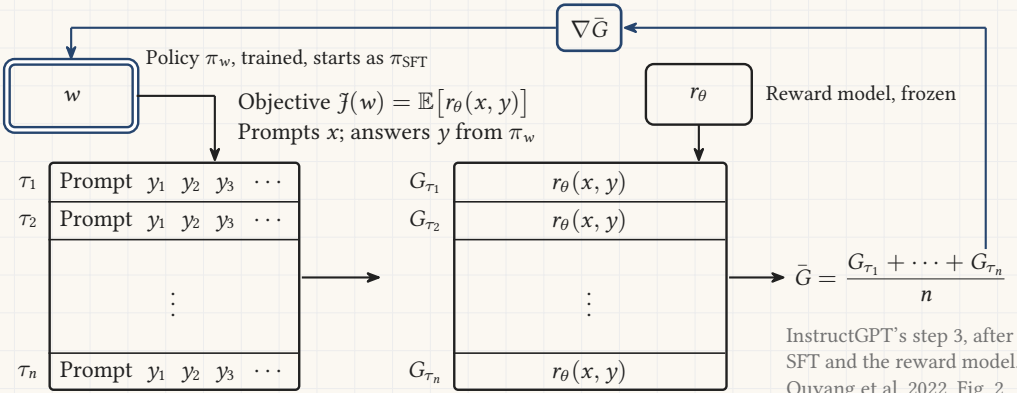
InstructGPT's step 3, after SFT and the reward model. Ouyang et al. 2022, Fig. 2

Reinforcement learning against the reward model



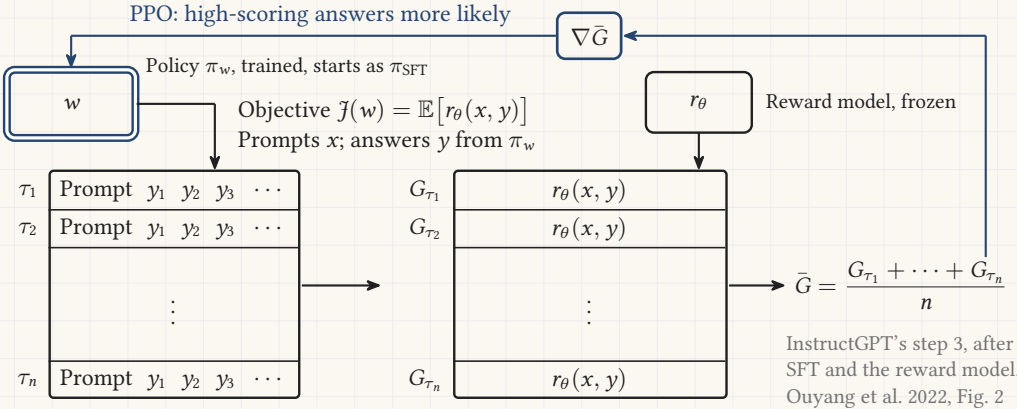
InstructGPT's step 3, after SFT and the reward model. Ouyang et al. 2022, Fig. 2

Reinforcement learning against the reward model



InstructGPT's step 3, after SFT and the reward model. Ouyang et al. 2022, Fig. 2

Reinforcement learning against the reward model

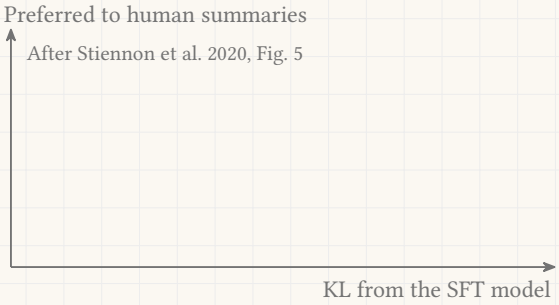


InstructGPT's step 3, after SFT and the reward model. Ouyang et al. 2022, Fig. 2

Maximising the reward model's score over-optimises

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big]$$

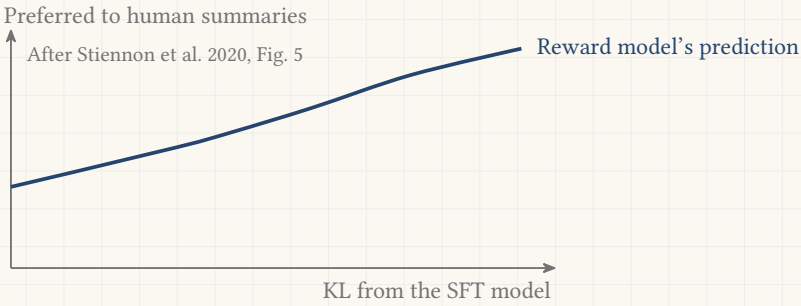
Prompts x; answers y from the policy π_w



Maximising the reward model's score over-optimises

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big]$$

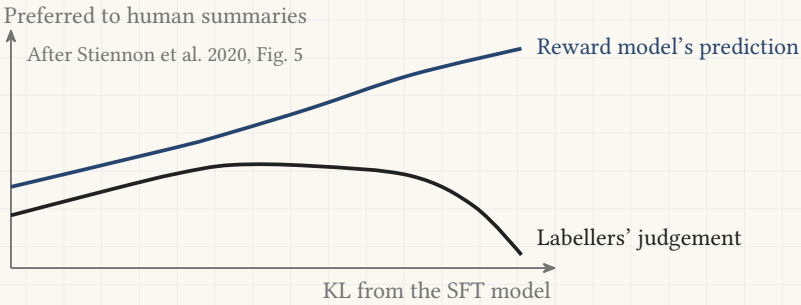
Prompts x; answers y from the policy π_w



Maximising the reward model's score over-optimises

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big]$$

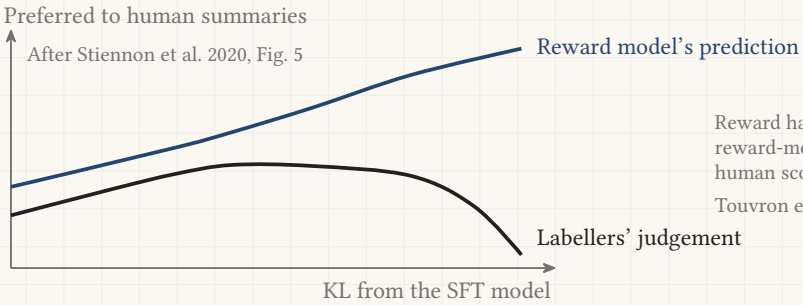
Prompts x; answers y from the policy π_w



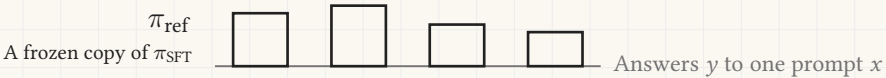
Maximising the reward model's score over-optimises

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big]$$

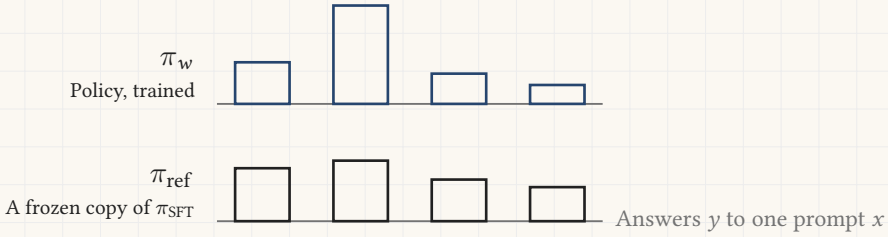
Prompts x; answers y from the policy π_w



The Kullback-Leibler divergence

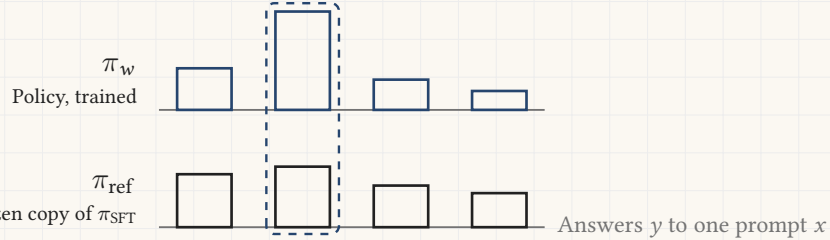


The Kullback-Leibler divergence



The Kullback-Leibler divergence

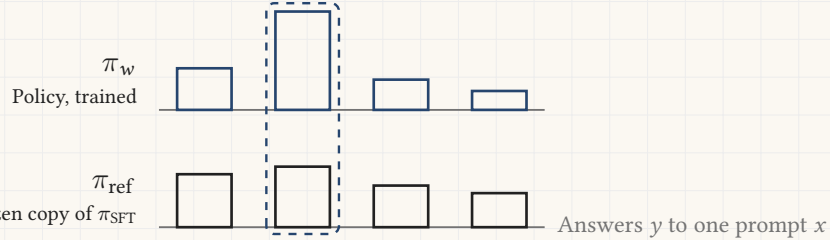
$$\underbrace{\log \frac{\pi_w(y \mid x)}{\pi_{\text{ref}}(y \mid x)}}_{\text{Log-ratio of one answer } y}$$



The Kullback-Leibler divergence

$$\mathrm{KL}\left(\pi_w \,\middle\|\, \pi_{\mathrm{ref}}\right) = \mathbb{E}\left[\underbrace{\log \frac{\pi_w(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)}}_{\text{Log-ratio of one answer } y}\right]$$

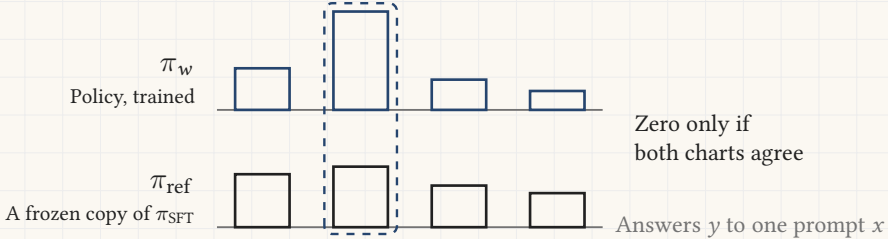
Over answers y drawn from π_w



The Kullback-Leibler divergence

$$\mathrm{KL}\left(\pi_w \,\middle\|\, \pi_{\mathrm{ref}}\right) = \mathbb{E}\left[\underbrace{\log \frac{\pi_w(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)}}_{\text{Log-ratio of one answer } y}\right] \geq 0$$

Over answers y drawn from π_w



The KL-regularised objective

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big]$$

Prompts x; answers y from the policy π_w

The KL-regularised objective

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big] - \beta \underbrace{\mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\text{ref}}\big)}_{\text{Drift from } \pi_{\text{ref}}}$$

Prompts x; answers y from the policy π_w

Reference π_{ref}: a frozen copy of π_{SFT}

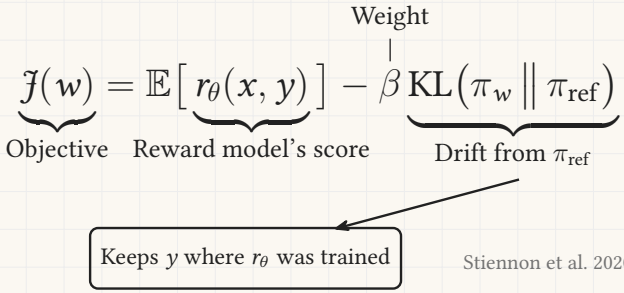
The KL-regularised objective

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big] - \underbrace{\beta}_{\text{Weight}}\underbrace{\mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\text{ref}}\big)}_{\text{Drift from } \pi_{\text{ref}}}$$

Prompts x; answers y from the policy π_w

Reference π_{ref}: a frozen copy of π_{SFT}

The KL-regularised objective

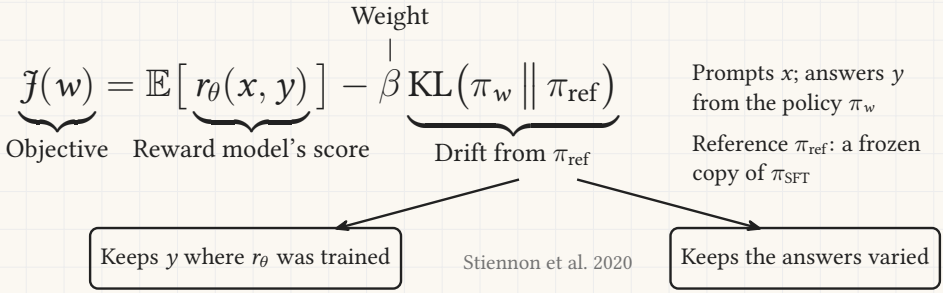


Stiennon et al. 2020

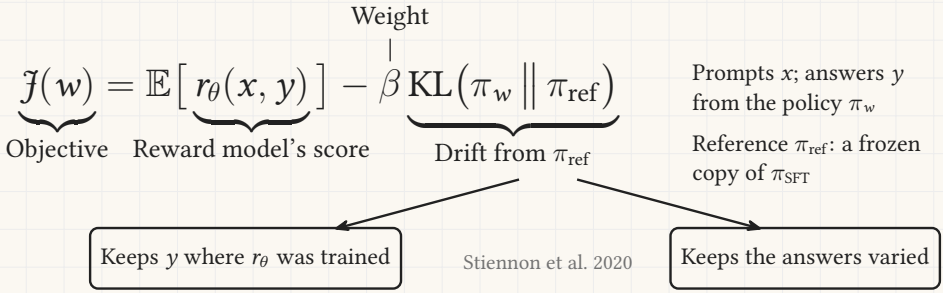
Prompts x; answers y from the policy π_w

Reference π_{ref}: a frozen copy of π_{SFT}

The KL-regularised objective



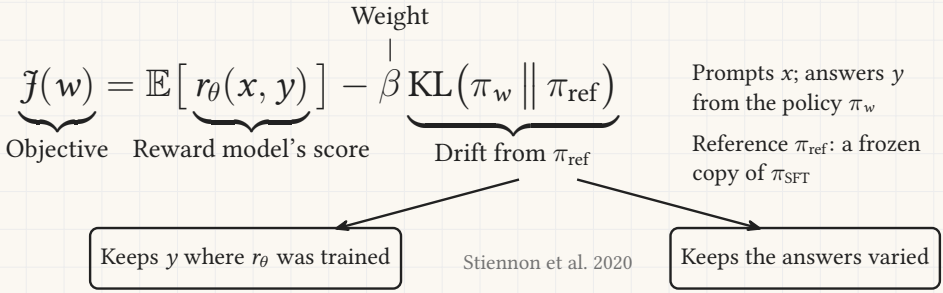
The KL-regularised objective



$\beta = 0.02$

InstructGPT

The KL-regularised objective



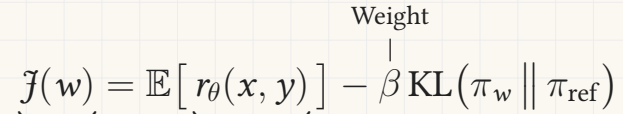
$\beta = 0.02$

InstructGPT

$\beta = 0.05$

Stiennon et al. 2020

The KL-regularised objective



Prompts x; answers y from the policy π_w

Reference π_{ref}: a frozen copy of π_{SFT}

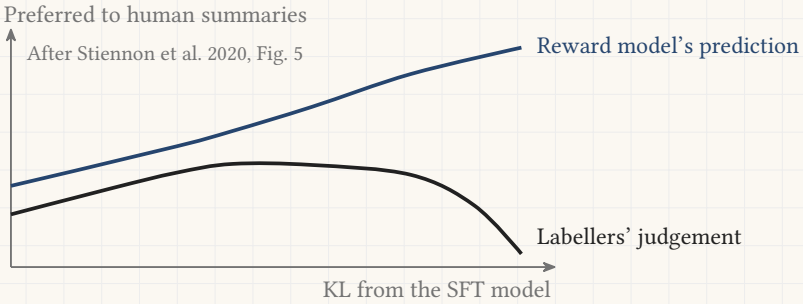
Keeps y where r_θ was trained

Stiennon et al. 2020

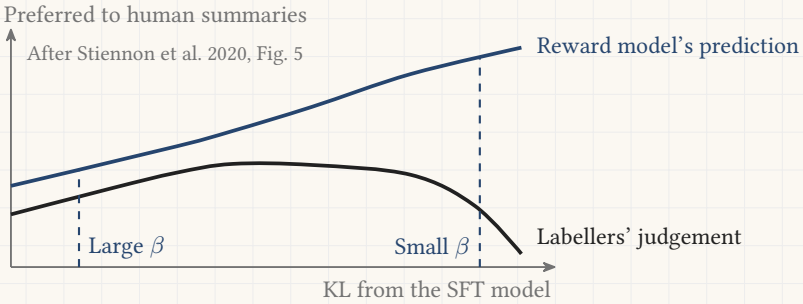
Keeps the answers varied

$\beta = 0.02$	$\beta = 0.05$	$\beta = 0.01$	$\beta = 0.005$
InstructGPT	Stiennon et al. 2020	Llama 2, 7B and 13B	Llama 2, 34B and 70B

The weight β trades reward against drift

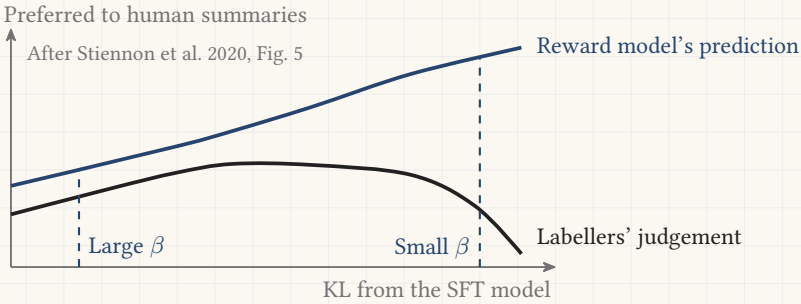


The weight β trades reward against drift



The weight β trades reward against drift

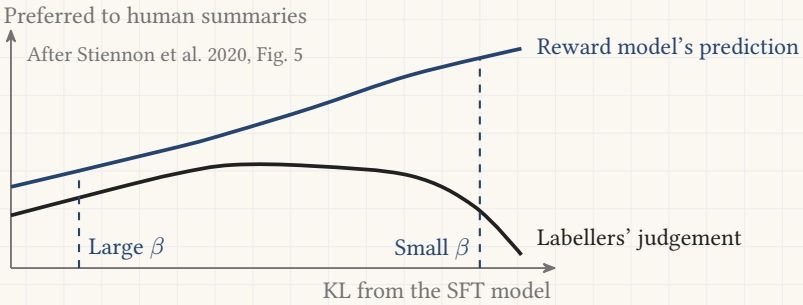
InstructGPT: best β near 0.01 to 0.02;
$\beta = 0$ and $\beta = 2$ both poor
Ouyang et al. 2022, App. E.7



The weight β trades reward against drift

InstructGPT: best β near 0.01 to 0.02;
$\beta = 0$ and $\beta = 2$ both poor
Ouyang et al. 2022, App. E.7

Penalty acts like early stopping
Gao et al. 2023

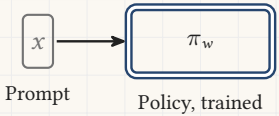


4

Section 4

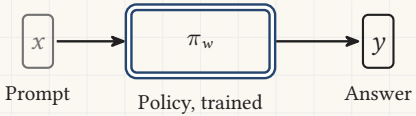
PPO against a reward model

Four networks



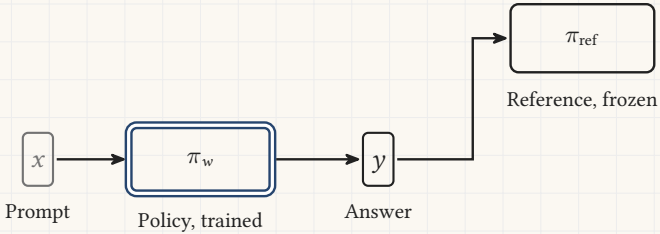
After Shao et al. 2024, Fig. 4

Four networks



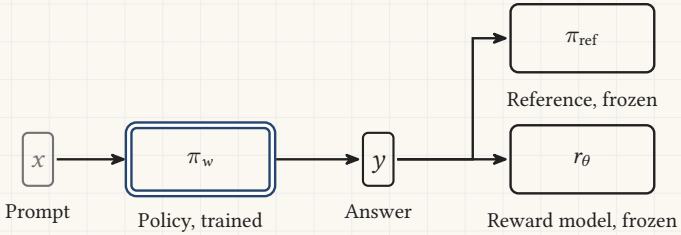
After Shao et al. 2024, Fig. 4

Four networks



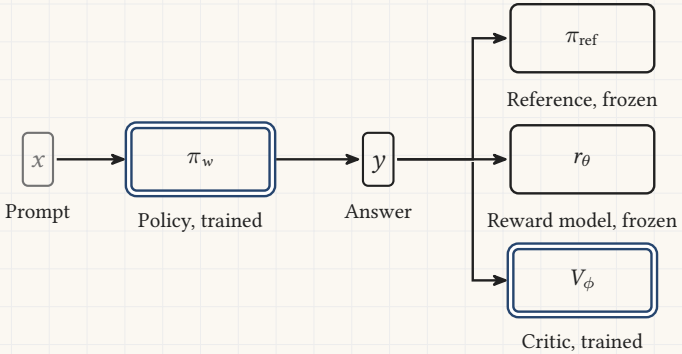
After Shao et al. 2024, Fig. 4

Four networks



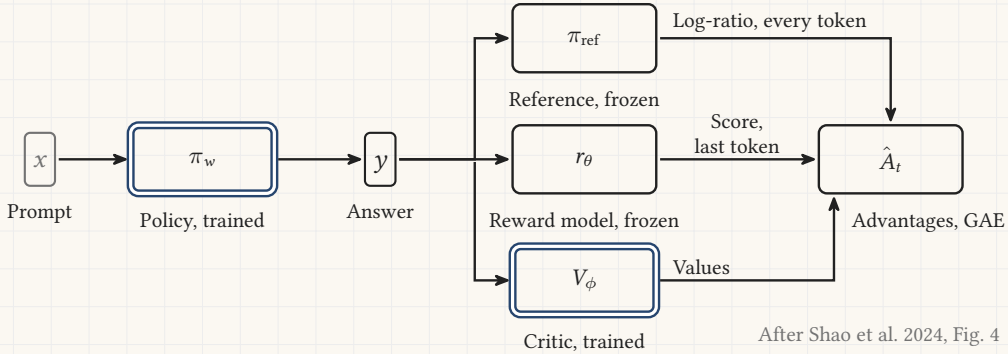
After Shao et al. 2024, Fig. 4

Four networks



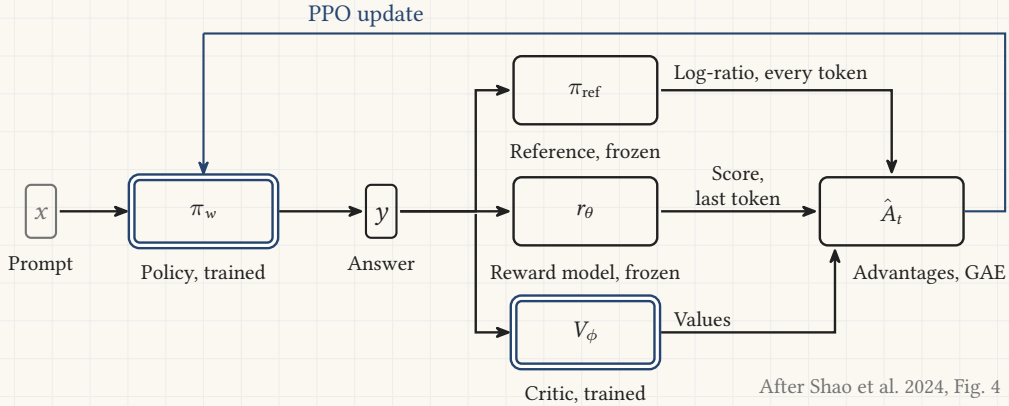
After Shao et al. 2024, Fig. 4

Four networks



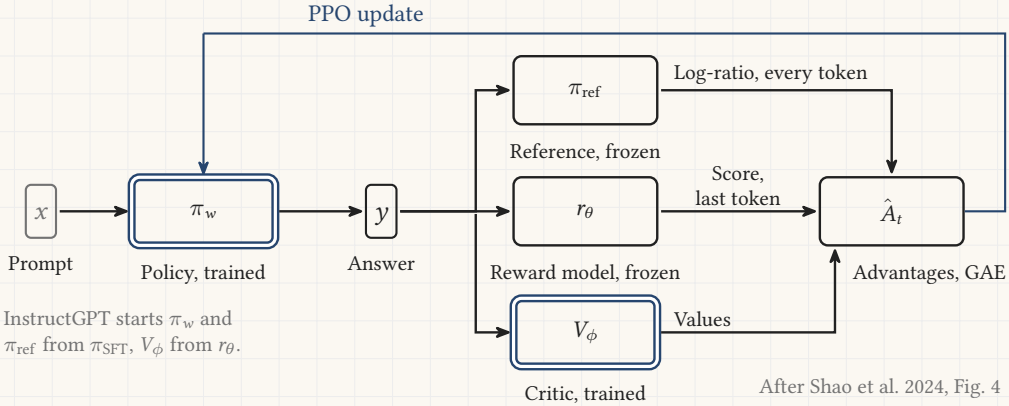
After Shao et al. 2024, Fig. 4

Four networks



After Shao et al. 2024, Fig. 4

Four networks



InstructGPT starts π_w and π_{ref} from π_{SFT}, V_ϕ from r_θ.

After Shao et al. 2024, Fig. 4

The alignment tax

$$\mathcal{J}(w) = \underbrace{\mathbb{E}\big[\, r_\theta(x, y) \,\big] - \beta\, \mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\mathrm{ref}}\big)}_{\text{KL-regularised objective}}$$

The alignment tax

$$\mathcal{J}(w) = \underbrace{\mathbb{E}\big[\, r_\theta(x, y) \,\big] - \beta\, \mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\text{ref}}\big)}_{\text{KL-regularised objective}}$$

PPO

SQuAD | DROP | HellaSwag | WMT, French to English

Below GPT-3

The alignment tax

$$\mathcal{J}(w) = \underbrace{\mathbb{E}\big[\, r_\theta(x, y) \,\big] - \beta\, \mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\text{ref}}\big)}_{\text{KL-regularised objective}} + \underbrace{\eta\, \mathbb{E}\big[\, \log \pi_w(\text{text}) \,\big]}_{\text{Pretraining data, weight } \eta}$$

PPO

SQuAD | DROP | HellaSwag | WMT, French to English

Below GPT-3

InstructGPT: $\eta = 27.8$

Ouyang et al. 2022, eq. (2), where it is γ

The alignment tax

$$\mathcal{J}(w) = \underbrace{\mathbb{E}\big[\, r_\theta(x, y) \,\big] - \beta\, \mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\text{ref}}\big)}_{\text{KL-regularised objective}} + \underbrace{\eta\, \mathbb{E}\big[\, \log \pi_w(\text{text}) \,\big]}_{\text{Pretraining data, weight } \eta}$$

PPO

SQuAD | DROP | HellaSwag | WMT, French to English

Below GPT-3

InstructGPT: $\eta = 27.8$

Ouyang et al. 2022, eq. (2), where it is γ

Regressions greatly reduced

The alignment tax

$$\mathcal{J}(w) = \underbrace{\mathbb{E}\big[\, r_\theta(x, y) \,\big] - \beta\, \mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\text{ref}}\big)}_{\text{KL-regularised objective}} + \underbrace{\eta\, \mathbb{E}\big[\, \log \pi_w(\text{text}) \,\big]}_{\text{Pretraining data, weight } \eta}$$

PPO

SQuAD | DROP | HellaSwag | WMT, French to English

Below GPT-3

InstructGPT: $\eta = 27.8$

Ouyang et al. 2022, eq. (2), where it is γ

Regressions greatly reduced

Labeller preference kept

References

Ralph A. Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika* 39(3/4), 324 to 345, 1952. doi.org/10.2307/2334029

Paul F. Christiano and others. Deep reinforcement learning from human preferences. *NeurIPS*, 2017. arxiv.org/abs/1706.03741

Nisan Stiennon and others. Learning to summarize from human feedback. *NeurIPS*, 2020. arxiv.org/abs/2009.01325

Long Ouyang and others. Training language models to follow instructions with human feedback. *NeurIPS*, 2022. arxiv.org/abs/2203.02155

Leo Gao, John Schulman and Jacob Hilton. Scaling laws for reward model overoptimization. *ICML*, 2023. arxiv.org/abs/2210.10760

Hugo Touvron and others. Llama 2: open foundation and fine-tuned chat models, 2023. arxiv.org/abs/2307.09288

Qwen Team (An Yang and others). Qwen2.5 technical report, 2024. arxiv.org/abs/2412.15115

Zhihong Shao and others. DeepSeekMath: pushing the limits of mathematical reasoning in open language models, 2024. arxiv.org/abs/2402.03300

Notation

x, y	A prompt, an answer	y^+, y^-	Chosen, rejected answer
y_t	Token t of y	c	Labeller's choice, 1 if y^+ won
π_w	The policy, trained; weights w	$\sigma(z)$	$1/(1+e^{-z})$
π_{SFT}	The model after SFT	β	Weight of the KL penalty
π_{ref}	A frozen copy of π_{SFT}	η	Weight of the pretraining term
r_θ	The reward model, weights θ	V_ϕ, $\hat{A}_t$	Critic, advantage estimate
$\mathbb{E}[\,\cdot\,]$	Expectation	$\mathrm{KL}(p \,\|\, q)$	Kullback-Leibler divergence